

EMIR BEKAR

Mechatronics Engineer · Game Developer · Content Creator

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SUMMARY

Mechatronics Engineering graduate (100% English curriculum) with strong mathematical and analytical foundations, award-winning senior design research, and two IEEE-published papers in multi-agent robotics and AI. Hands-on experience building mobile games in Unity and managing an active 12,000+ subscriber YouTube community with first-hand insight into player retention and content iteration. Passionate about mobile casual games and eager to drive engaging product features and analytics at Peak.

EDUCATION

Istanbul Bilgi University Jul 2026
Bachelor of Science in Mechatronics Engineering (100% English Curriculum) Istanbul, Turkey

- 🏆 **Best Senior Design Project Award:** "Adaptive Formation Control for Collaborative Swarm Robots" (Presented at HORA Congress 2026, published on IEEE Xplore).

FEATURED PROJECTS

Match 3D Mobile Game Unity 6 · C# · Blender · URP

- Developed a hybrid-casual 3D matching mobile game with physics-based matching and zero-GC object pooling for stable 60 FPS.
- Designed core retention and monetization meta-game loops: Daily Rewards, Lucky Spin wheel, and in-game economy store.
- Modeled and textured modular 3D assets in Blender for mobile visual clarity. [\[GitHub\]](#) | [Video Demo ↗](#)

AC-130 Gunship Simulation (In Development) Unity 6 · C# · Custom Shaders

- Close air support tactical simulation featuring custom FLIR thermal vision shaders, dynamic terrain destruction, and avionics HUD telemetry. [\[GitHub\]](#) | [Video Demo ↗](#)

Forza Horizon 6 Mobile Cockpit Node.js · UDP · Mobile · Realtime

- Wireless 60Hz real-time telemetry racing dashboard converting mobile devices into responsive UDP telemetry instruments. [\[GitHub\]](#)

LAN Drop Python · FastAPI · Streaming · Zero-Config

- High-throughput, zero-config local Wi-Fi file transfer tool with 8MB stream chunking and ASCII QR terminal connectivity. [\[GitHub\]](#)

PUBLICATIONS (IEEE XPLORE · HORA CONGRESS 2026)

- Adaptive Formation Control for Collaborative Swarm Robots (IEEE 2026):** Leader-follower formation switching via FSM and fuzzy-logic ACC control eliminating stop-and-go oscillations, validated in Webots simulation. [\[IEEE Xplore #11537161 ↗\]](#)
- Multi Agent Reinforcement Learning Based Swarm Navigation (IEEE 2026):** Autonomous swarm coordination and collision-free path optimization using Multi-Agent Reinforcement Learning (MARL). [\[IEEE Xplore #11537199 ↗\]](#)

COMMUNITY & CONTENT CREATION

Emiukob – YouTube Creator & Mod Developer 12,000+ Subscribers · youtube.com/@emiukob

- Built a gaming and tech community to 12,000+ subscribers; analyzed audience retention curves, drop-off rates, and CTR to iteratively optimize content pacing.
- Developed and published "Hologram Projector" 3D hologram Minecraft mod in Java on CurseForge. [\[CurseForge ↗\]](#)

SKILLS & CERTIFICATIONS

Game & Product: Mobile casual / hybrid-casual games, 3D match mechanics, level pacing, FTUE, player retention & meta-game loops

Technical & Tools: Unity 6 (C#), Python, JavaScript, Java, Node.js, Blender, Web Audio API, Git/GitHub, Webots Simulation

Analytical: Data interpretation, A/B testing mindset, mathematical modeling, fuzzy logic, reinforcement learning

Certifications: Basic Python for Business Analytics (Arizona State University) | UAV 0/1 Pilot License (SHGM)